

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	BS-EE
Program Name	BS in Electrical Engineering
Assessment Year	2025
Program Assessment Coordinator	Erdal Oruklu
Year of Current Assessment Plan	AY2024
No Assessment This Year per Plan	☐
LO(s) Assessed	LO(4), LO(5), LO(6), LO(7)
Presented at program authority meeting date: <i>Please attach minutes</i>	10/14/2025

Methodology

For each LO and artifact assessed, please describe methodology used, and attach (as supplemental files during report submission) or insert here the rubric used. At minimum this should contain:

- a description of how the artifact is concordant with the LO
- a rubric describing the method used to place individual student in at least 3 assessment categories; {**D** Does not meet expectations, **M** Meets expectations **M**, **P** Proficient}

The student learning goals assessed this cycle are that, at the time of graduation, students have:

- [4] *An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.*
- [5] *An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.*
- [6] *An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.*
- [7] *An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.*

(Please see the attached rubric files for each LO).

- **LO(4)** has 3 Performance Indicators in 4 assessment categories (Beginning, Developing, Proficient and Exemplary)
LO(4) assessment is done in Spring 2025 with the following courses and instructors. Homework or exam problems are collected as student artifacts in these courses.

ECE 430 (S25): T. Wong

ECE 481 (S25): J. Kim

Total sample size is 27 (Three performance indicators for each of 9 students). Artifacts were assessed during the semester in which the homework assignment or examination was given (S25).

- **LO(5)** has 3 Performance Indicators in 4 assessment categories (Beginning, Developing, Proficient and Exemplary)
LO(5) assessment is done in Fall 2024 and Spring 2025 with the following courses and instructors. Homework or exam problems are collected as student artifacts in these courses.

ECE 411 (F24): F. Shang

ECE 441 (F24): J. Saniie

ECE 407 (S25): L. Cai

ECE 412 (S25): F. Shang

ECE 441 (S25): J. Saniie

Total sample size is 165 (Three performance indicators for each of 55 students). Artifacts were assessed during the semester in which the homework assignment or examination was given (F24 or S25).

- **LO(6)** has 4 Performance Indicators in 4 assessment categories (Beginning, Developing, Proficient and Exemplary)
LO(6) assessment is done in Fall 2024 and Spring 2025 with the following courses and instructors. Homework or exam problems are collected as student artifacts in these courses.

ECE 411 (F24): F. Shang

ECE 441 (F24): J. Saniie

ECE 407 (S25): L. Cai

ECE 412 (S25): F. Shang

ECE 441 (S25): J. Saniie

Total sample size is 220 (Four performance indicators for each of 55 students). Artifacts were assessed during the semester in which the homework assignment or examination was given (F24 or S25).

- **LO(7)** has 3 Performance Indicators in 4 assessment categories (Beginning, Developing, Proficient and Exemplary)
LO(7) assessment is done in Fall 2024 and Spring 2025 with the following courses and instructors. Homework or exam problems are collected as student artifacts in these courses.

ECE 408 (F24): L. Cai

ECE 425 (F24): S. Liao

ECE 443 (F24): J. Wang

ECE 449 (F24): W.-J. Yi

ECE 485 (F24): S. Borkar

ECE 408 (S25): L. Cai

ECE 420 (S25): H. Shanechi

ECE 438 (S25): D. Veselinovic

ECE 448 (S25): W.-J. Yi

ECE 481 (S25): J. Kim

Total sample size is 189 (Three performance indicators for each of 63 students). Artifacts were assessed during the semester in which the homework assignment or examination was given (F24 or S25).

Results

Artifact samples

Artifact samples are attached for each LO.

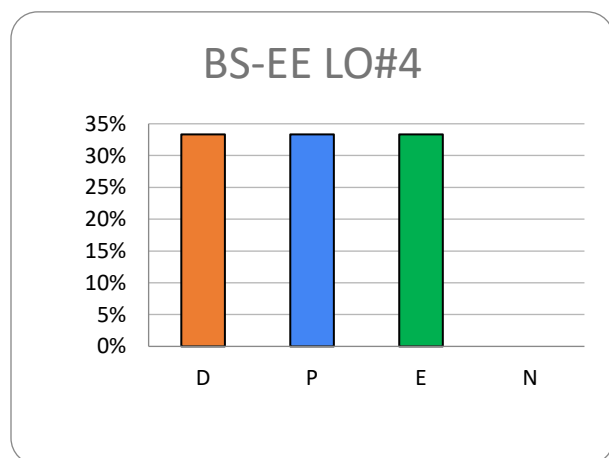
Data table

For each LO and provide a table pasted in from the downloadable [provided template](#), placing each student into the assessment categories (minimally {D, M, P}). You should collect and analyze data linked to student A#s locally, but please redact any personally identifiable information, PII (including names or A#s) in tables presented in this report. Provide a summary graph from the template presenting this distribution. You should also present additional analysis as deemed appropriate by the results seen, and the needs of the program.

Assessment data is provided in the attached report template. A summary of the results is shown in the graphs below.

LO(4): An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

66% of students achieve a “Proficient” or “Exemplary” score:



D: Does not meet expectations

P: Proficient

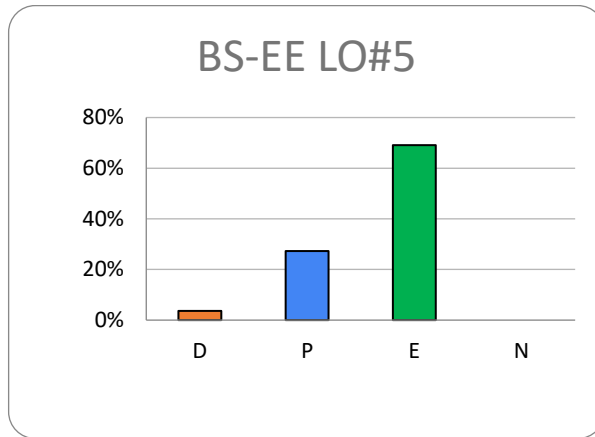
E: Exemplary

N: Not assessed

LO(4) PERFORMANCE INDICATORS	PERCENTAGE SCORING “PROFICIENT” OR “EXEMPLARY”
PI 1: Identify the global, economic, environmental, and societal context of an engineering situation.	66.7%
PI 2: Describe ethical and professional responsibilities related to an engineering project.	66.7%
PI 3: Explain the impact of engineering decisions in a global, economic, environmental, and societal context.	77.8%

LO(5): An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

96% of students achieve a “Proficient” or “Exemplary” score:



D: Does not meet expectations

P: Proficient

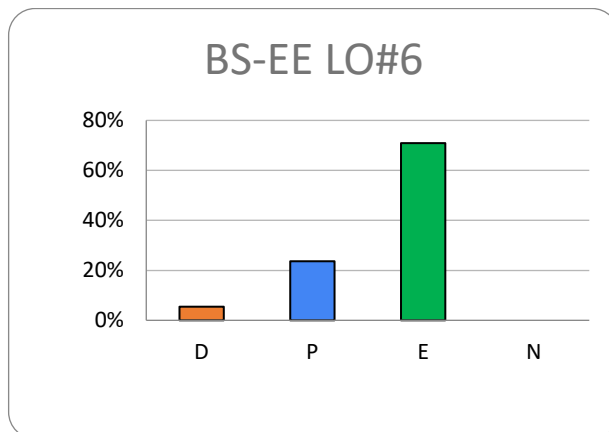
E: Exemplary

N: Not assessed

LO(5) PERFORMANCE INDICATORS	PERCENTAGE SCORING “PROFICIENT” OR “EXEMPLARY”
PI 1: Establish a collaborative and inclusive environment (Teamwork).	96%
PI 2: Fulfill individual responsibilities and contribute to the team’s success. (Individual accountability)	96%
PI 3: Define team goals and deadlines, plan tasks, organize & facilitate effective team meetings. (Project management)	96%

LO(6): An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

95% of students achieve a “Proficient” or “Exemplary” score:



D: Does not meet expectations

P: Proficient

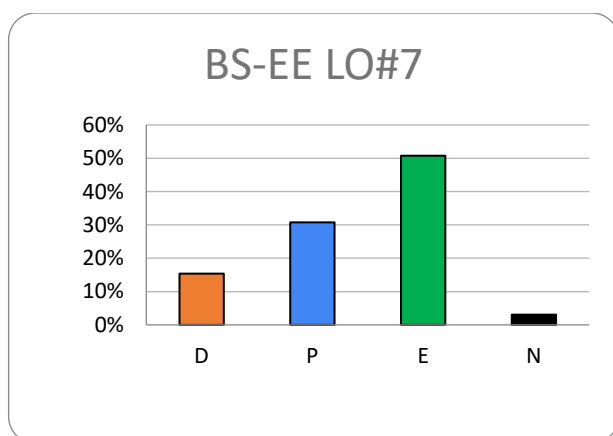
E: Exemplary

N: Not assessed

LO(6) PERFORMANCE INDICATORS	PERCENTAGE SCORING "PROFICIENT" OR "EXEMPLARY"
PI 1: Design an Experiment Plan (How to answer the Driving Question?)	96%
PI 2: Acquire data on appropriate variables	89%
PI 3: Interpret experimental data and results with respect to appropriate theoretical models	89%
PI 4: Explain observed differences between model and experiment (bad model, bad measurements, noise, etc.) and draw conclusions	95%

LO(7): An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

84% of students achieve a "Proficient" or "Exemplary" score:



D: Does not meet expectations

P: Proficient

E: Exemplary

N: Not assessed

LO(7) PERFORMANCE INDICATORS	PERCENTAGE SCORING "PROFICIENT" OR "EXEMPLARY"
PI 1: Identify necessary techniques, skills and tools for a new situation (research)	84%
PI 2: Explain the use of the new techniques, skills and tools (acquisition)	81%
PI 3: Apply the new techniques, skills and tools to the given situation	83%

Description of findings

Summarize the discussion that occurred within the program authority when this was presented. Provide a short narrative interpreting the findings. Are there any areas of concern?

Artifacts are evaluated with respect to several performance indicators, with a rating of (1) beginning, (2) developing, (3) proficient, or (4) exemplary. Each student outcome has its own unique set of performance indicators, and the performance indicators are evaluated using a standard rubric adopted by the ECE Department. Assessment results are summarized by displaying the percentage of artifacts from 400-level courses that received either a (3) proficient or (4) exemplary evaluation. This percentage is shown for each performance indicator.

The ECE Department has examined the assessment results and noted that **Student Learning Outcome LO (4) was assessed at 66%, LO (5) assessed at 96%, LO (5) assessed at 95%, and LO (7) assessed at 84%.** Therefore, LO(5), LO(6) and LO(7) are all above the desired 75% threshold as defined by the ECE Department.

LO (4) was assessed 66%, which is below the target 75%. In addition, the number of students assessed for LO(4) was limited to 9.

Continuous Improvement, CI, plans

The purpose of assessment is to improve the quality of our educational programs. The most important part of the assessment cycle is how we use the data to improve our programs. Examine the data provided above describing student achievement, consider the root cause of any deficits, and if areas for possible improvement emerge, describe plans to achieve this improvement.

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)?

☐ no previous CI plans available

LO(4), LO(5), LO(6) and LO(7) were last assessed in the AY2022-23 cycle. For AY2022-23, (LO) (5) was assessed at 100%, SO (6) assessed at 92%, and SO (7) assessed at 89%. They were all above the desired 75% threshold.

LO (4) assessment was limited to one course only and all three performance indicators were below the 75% threshold.

CI plan in Ay2022-2023 recommended the following actions to be implemented by the ECE Department:

- ECE Undergraduate Program Committee (UPC) recommends additional courses within the EE major to cover the SO (4) and SO (7). UPC will contact the respective course instructors for the necessary modifications. The new assessment cycle will target AY 24/25.

Specific modification to courses or assignments or curriculum.	Name and title of person responsible for implementing the changes.	Date by which changes will be in place.	Intended result
ECE 218: New labs introduced for FPGA design with CAD tools	S. Borkar	F23	SO(7) PI3 will be improved.
UPC will contact course instructors for possible modifications to emphasize societal and ethical challenges in engineering - SO(4)	J. Kim (ECE 481) Y. Cheng (ECE 444) J. Saniie (ECE 222 and ECE 442) T. Wong (ECE 430)	Spring 2025	Improve SO(4) performance

In AY2024-2025, LO(5), LO(6) and LO(7) continue to be assessed well above the 75% threshold. LO(7) PI3 (“Apply the new techniques, skills, and tools to the given situation”) has improved to 83% compared to 67% in AY2022-2023. Additional lab experiments and tools introduced in ECE218 may have contributed to this improvement.

LO(4) assessment continues to be below the threshold (66%). Sample data is still low (only 9 students) despite introducing additional courses such as ECE 442 for assessment.

This cycle's CI plan

Describe any CI plans developed from this cycle's data, with

- Specific curricular or instructional changes in response to this assessment data,
- A timeline for implementation,
- As well a description of who will participate in and manage these plans

☐ After consideration and presentation to the program authority, no CI plans are recommended.

The ECE department will continue to monitor and evaluate the course assignments of student outcomes in order to establish a balanced data collection procedure. The ECE Department will seek collaboration with the Department of Humanities and "Center for the Study of Ethics in Professions" for improved assessment process of LO(4).

Two courses will be modified to improve the LO(7) outcomes, including ECE 420 and ECE 485.

Specific modification to courses or assignments or curriculum.	Name and title of person responsible for implementing the changes.	Date by which changes will be in place.	Intended result
LO(4) assessment will be implemented with the help of Illinois Tech's "Center for the Study of Ethics in Professions"	ECE 100 (C. Zhou), ECE 441 (J. Saniie) Elisabeth Hildt and Kelly Laas (Ethics Center)	Fall 2025/Spring 2026	Improved LO(4) results with: More reliable data, more robust assignments that teach students ethics, less variation between assessments.
ECE 420: Add stochastic state estimation to chapter 5.	H. Shanechi	Spring 2026	LO(7) – all three PIs
ECE 485: Provide a training session on VHDL	S. Borkar	Fall 2025	LO(7)- PI1 and PI3 (new tolos)